AMC 10 Crash Course · Test T2 · Geometry

10 problems · 30 minutes · no calculator · mastery bar 7 of 10

Name: ______________________ Date: ______________________

1. The exterior angle of a regular polygon is $30°$. How many sides does the polygon have?

(A) 8 **(B)** 9 **(C)** 10 **(D)** 12 **(E)** 15

2. The angles of a triangle are in the ratio $3 : 4 : 5$. What is the largest angle (in degrees)?

(A) 60 **(B)** 70 **(C)** 75 **(D)** 80 **(E)** 90

3. A right triangle has a $30°$ acute angle and hypotenuse 12. Find the leg opposite the $60°$ angle.

(A) $3\sqrt{3}$ **(B)** 6 **(C)** $6\sqrt{2}$ **(D)** 9 **(E)** $6\sqrt{3}$

4. The legs of a right triangle are 6 and 8. Find the altitude to the hypotenuse.

(A) 3.6 **(B)** 4.8 **(C)** 5 **(D)** 6.4 **(E)** 7

5. Find the area of the triangle with sides 10, 10, and 12.

(A) 36 **(B)** 44 **(C)** 48 **(D)** 54 **(E)** 60

6. In triangle ABC, point M lies on side AB and point N on side AC, with $AM : AB = 2 : 3$ and $AN : AC = 3 : 4$. The area of triangle ABC is 96. Find the area of quadrilateral $BMNC$.

(A) 40 **(B)** 44 **(C)** 48 **(D)** 52 **(E)** 56

7. Point P is at distance 17 from the center of a circle of radius 8. Find the length of the tangent from P to the circle.

(A) 9 **(B)** 12 **(C)** 13 **(D)** 15 **(E)** 25

8. Triangle ABC has sides $AB = 13$, $BC = 14$, $CA = 15$. The incircle touches side BC at point T. Find BT.

(A) 5 **(B)** 6 **(C)** 7 **(D)** 8 **(E)** 9

9. Find the area of the triangle with vertices $(0, 0)$, $(10, 4)$, and $(2, 6)$.

(A) 18 **(B)** 20 **(C)** 22 **(D)** 24 **(E)** 26

10. Find the space diagonal of a 4-by-4-by-7 rectangular box.

(A) 9 **(B)** 10 **(C)** 11 **(D)** 12 **(E)** 15

Answer Key and Review

Parent page: cut off or do not print for the student. Map misses to lessons: 1–2 → 2.1, 3–4 → 2.2, 5–6 → 2.3, 7–8 → 2.4, 9 → 2.5, 10 → 2.6.

1	2	3	4	5	6	7	8	9	10
D	**C**	**E**	**B**	**C**	**C**	**D**	**B**	**E**	**A**

1. Exterior angles sum to 360°: $n = 360/30 = 12$.

2. One part = 180/12 = 15°: the largest angle is $5 \cdot 15 = 75°$.

3. The pattern $x : x\sqrt{3} : 2x$: $x = 6$, and $6\sqrt{3}$ lies opposite 60°.

4. Hypotenuse 10, altitude = $6 \cdot 8/10 = 4.8$ (while 3.6 and 6.4 are the hypotenuse segments — traps).

5. The altitude to the side of 12: $\sqrt{(100 - 36)} = 8$, so the area is $12 \cdot 8/2 = 48$.

6. $[AMN] = (2/3)\cdot(3/4)\cdot 96 = 48$; quadrilateral: $96 - 48 = 48$. With a shared angle, the area ratio is the product of the side ratios.

7. $\sqrt{(17^2 - 8^2)} = \sqrt{225} = 15$ (the 8-15-17 triple).

8. Tangent segments: $x+y = 13$, $y+z = 14$, $x+z = 15$; add and subtract: $y = BT = 6$ (= $s - CA = 21 - 15$).

9. Shoelace with the zero vertex first: $|10 \cdot 6 - 2 \cdot 4|/2 = 52/2 = 26$.

10. $\sqrt{(16 + 16 + 49)} = \sqrt{81} = 9$.